Texas A&M Science Olympiad Society Div B Invitational

January 17th, 2026

Water Quality

Event Parameters:

- Each participant may bring one 8.5” x 11” sheet of paper that may contain information on both sides in any form and from any source without any affixed labels.
- Each team may bring one studnet built a salinometer/hydrometer for testing
- Participants may bring non-programmable, non-graphing calculators.
- There are 58 questions worth 170 points
    - All questions have been labelled point values, tie-breakers are decided by the testing coordinator

Directions:

- DO NOT WRITE ON THE TEST
- Write answers on the answer sheet provided
- You have 50 minutes
- Participants must wear eye protection during the Salinometer Testing

1. What percentage of the freshwater on Earth is groundwater, and what percentage is surface water (lakes, rivers, wetlands, etc)? (2pts)

    a. Ground water: _____ (%)

    b. Fresh water: _____ (%)

2. Fill in the blank:

    Only about _____ percent of the energy from one tropic level is transferred to to the next. (1pts)

    Match the terms to the definitions: (4pts)

3. _____ Total quantity of dissolved salts in water — A. Ph

4. _____ Excess nutrients lead to algae blooms, which deplete oxygen levels when they decompose — B. Salinity

5. _____ A scale that measure the acidity or alkalinity of a liquid — C. Turbidity

6. _____ The measure of water clarity, or the amount of suspended solids in the water — D. Eutrophication

7. What is the most common pollutant in rivers, streams, lakes, and reservoirs? (1pts)
    A) Oil and gasoline
    B) Raw sewage
    C) Pesticides and herbicides
    D) Sediment

8. What is the primary cause of sedimentation pollution? (1pts)
    A) Disease-causing agents
    B) Fertilzers
    C) Erosion
    D) Toxic compounds

9. Label each part of the water cycle: (4pts)

a. _____

b. _____

c. _____

d. _____

10. Order the hierarchy of biological organizations from least to most complex. (6pts)

Organism Abiotic Habitat Population Energy
Community Climate Ecosystem Biome

a. _____

b. _____

c. _____

d. _____

e. _____

f. _____

11. What phase during wastewater treatment is water mixed with air and sludge filled with bacteria? Which of these bacteria is used in this phase? (2pts)
    A) Screening; *Pseudomonas aeruginosa*
    B) Aeration; Aerobic bacteria
    C) Disinfection; *E.coli*
    D) Sedimentation; Anaerobic bacteria

12. Which of the following processes describes the movement of water through the Earth's systems? (1pts)
    A) Nutrient cycling
    B) Population dynamics
    C) The water cycle
    D) Community interactions

13. Fill in the blank:

The process of cleaning and disinfecting water to make it safe for human consumption is called _____ (1pts)

14. In a freshwater ecosystem, which organism group occupies the lowest trophic level, producing its own food? (1pts)
    A) Phytoplankton
    B) Zooplankton
    C) Fungi
    D) Moths

15. Identify the organism. (1pts)

    _____

16. Based on the insect's size and structure, what might its primary role be in an aquatic food web? (1pts)

    _____

17. Observe the short forelegs and mouthparts: what feeding method do these features suggest? (1pts)

    _____

18. This aquatic insect has a sharp, piercing mouthpart called a proboscis. What is its primary function? (1pts)
    A) Sucking nectar from aquatic flowers
    B) Filtering algae and plankton from the water
    C) Cutting through tough aquatic vegetation
    D) Injecting venom into prey and consuming their fluids

19. Identify the organism. (1pts)

    _____

20. The prominent, feathery antennae on this insect indicate which of the following? (1pts)
    A) The insect is a female that needs to locate a place to lay eggs.
    B) The insect is a beneficial pollinator that senses nectar sources.
    C) The insect uses these antennae primarily for navigating air currents during flight.

D) The insect is a male that uses these structures to detect the flight tone (sound) of females for mating.

21. In which environment does this insect spend the majority of its life cycle (the larval stage)? (1pts)
    A) Underground in damp soil and leaf litter.
    B) Inside the stems of aquatic vegetation.
    C) At the bottom of a water body (pond, lake, or stream).
    D) In the sap of trees found in wetland areas.

22. In many aquatic ecosystems, this organism's larvae play a crucial role as _____ of water quality, as their pressese or absence can indicate certain environmental conditions. (1pts)

23. Identify the organism (1pts)

    _____

24. This insect is an important bioindicator species in aquatic ecosystems. Its presence in a stream is typically a sign of what water quality condition? (1pts)
    A) High levels of acidity of pollution
    B) High levels of dissolved oxygen and minimal pollution
    C) High water temperatures
    D) Low levels of dissolved oxygen

25. The aquatic immature stage of this insect, which lives for one to four years, is called a _____ (1pts)

26. Identify the organism (1pts)

    _____

27. The presence of a diverse population of pollution-sensitive organisms, like the creature pictured, usually indicates (1pts)
    A) real-time chemical concentrations
    B) localized pH fluctuations
    C) short-term changes in water temperature
    D) long-term ecosystem health

28. The type of organism that is shown in the image (1pts)
    A) is an arachnid
    B) is used primarily to measure the pH levels

C) is often found in polluted water environments
D) is classified as an insect

29. During the six-legged _____ stage, the mite lives as an ectoparastie, often attaching to aquatic insects like dragonflies or water boatmen to feed on their fluids and travel to new habitats. (1pts)

Students are analyzing two different industrial discharge points that feed into a local river. They measure the hydrogen ion concentration ([H^+]) for samples taken at each site.

- Site A: [H^+] concentration is 1.0 * 10^-6 moles/Liter
- Site B: [H^+] concentration is 1.0 * 10^-4 moles/Liter

30. Calculate the pH of site A: _____ (2pts)

31. Calculate the pH of site B: _____ (2pts)

Fill in the blanks:

32. Site _____ is more acidic than Site _____ by _____ times. (3pts)

The bar chart displays data collected during a water treatment process, tracking changes in Tubidity (a measure of water cloudiness) and Dissolved Oxygen (DO) levels.

33. Calculate the percentage reduction in turbidity between raw and tap water _____ % (round to the nearest whole number) (2pts)

34. Calculate the increase in dissolved oxygen concentration between raw and tap water _____ % (round to nearest whole number) (2pts)

35. What two processes explain the dissolved oxygen change throughout the treatment? (3pts)

    a. _____

    b. _____

Texas A&M Invitational - Repaired practice edition

Division B | Water Quality | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

repair-case. (4 points) Authored written substitute for unavailable source material: Explain the evidence supporting your answer, including the calculation or comparison.

Unit: Calibrate a Division B salinometer.
Worked-case data: 4.0% standard: Immersion is 9.8 cm. 6.0% standard: Immersion is 9.6 cm. Unknown: Immersion is 9.7 cm at the same temperature.
Separate investigation observations:
4.0% standard: Immersion is 9.8 cm.
6.0% standard: Immersion is 9.6 cm.
Unknown: Immersion is 9.7 cm at the same temperature.

Answer: _____
_____
_____